Research on Planning Pathways for Low-Carbon Rural Construction —Taking Three Typical Villages in Zhejiang Province as Examples

Abstract: Against the backdrop of actively implementing China's carbon peak and carbon neutrality policies, high-quality development is being further advanced in the field of rural development. However, systematic low-carbon planning guidance remains insufficient in current rural construction practices, and many regions still rely on traditional high-carbon and extensive planning approaches.

This study establishes a planning cognition framework aimed at achieving low carbon village construction. Specifically, it elaborates relevant planning and design methods as well as implementation pathways from six dimensions: overall planning layout, industrial planning, transportation planning, energy and infrastructure planning, green space system planning, and architectural design guidelines. Subsequently, three representative villages are introduced as empirical case studies: Xinjian Village in Ganlan Town in Dinghai District of Zhoushan City, Lotus International Future Community in Lotus Village of Qujiang District of Quzhou City, and Shatan Village in Yutou Township of Huangyan District of Taizhou City. Through in-depth investigation of these cases, the study extracts and summarizes planning pathways for low-carbon rural development, with the aim of providing guidance for the widespread rural construction practices currently taking place across China.

Key Words: low carbon village; rural revitalization; planning path; Xinjian Village in Ganlan Town; Lotus International Future Community in Lotus Town; Shatan Village in Yutou Township

1 Introduction

Since the Industrial Revolution, economic development has been accompanied by a rapid increase in greenhouse gas emissions, dominated by CO2, and the world has made numerous efforts to combat climate change. In 2020, China announced that it would strive to peak its carbon dioxide emissions before 2030 and achieve carbon neutrality before 2060("dual carbon" goals). Advancing carbon peaking and carbon neutrality in depth requires integrating the dual carbon goals into the target system of high-quality development.

In the process of jointly advancing the "dual carbon" goals and the rural revitalization strategy, rural construction is facing unprecedented challenges. First, rapid industrialization and urbanization have brought about tremendous changes in both the industrial structure and the employment structure of rural areas[1]; high-carbon-emission industries in developed regions have shown a tendency to relocate to less developed areas such as suburbs and the countryside, and the high-carbon functions of cities have repeatedly penetrated into rural areas[2]. Second, changes in lifestyle have led to increased carbon emissions: the natural wisdom of traditional villages—building in harmony with the terrain and adapting to local climate[3]—has been indiscriminately abandoned in the course of blindly copying urban development models, and the spread of motorized transportation and commercial energy has caused the energy intensity of rural housing to exceed that of urban areas in some respects. Finally, population growth and a development mode characterized by high pollution and high emissions have posed significant challenges to the carrying capacity of resources and the environment, such as soil and water bodies; ecosystem functions have declined, and carbon sink functions can hardly be brought into full play.

Implementing the dual carbon goals requires organic integration and coordinated advancement with the rural revitalization strategy. Rural areas are rich in ecological resources and enjoy advantages unmatched by cities in performing carbon sink functions; the overall layout of traditional villages embodies the natural wisdom of our ancestors and provides a solid foundation for reducing carbon emissions and mitigating environmental impacts. However, the preparation of some village plans in China currently lacks a systematic, low-carbon-oriented understanding and dedicated pathway design, making it difficult to achieve the goals of carbon reduction and carbon sequestration. Therefore, there is an urgent need to distill low-carbon rural planning pathways that are both universally applicable and operable.

The key questions of this study focus on three aspects: first, what the goals of low-carbon rural construction are, how low-carbon goals can be achieved at different spatial levels, and through which planning pathways they can be realized; second, which planning pathways have already been applied in low-carbon rural construction practice, and how effective they have been; and third, what lessons can be drawn from existing low-carbon rural construction practices, and how these lessons can guide low-carbon rural construction in China.

2 Theoretical and Practical Progress of Low-Carbon Villages

2.1 Theoretical Development of Low-Carbon Villages

Since the 20th century, with the introduction of concepts such as "low-carbon society"[4] and "low-carbon city"[5], rural areas have also begun to be regarded as important spaces for low-carbon transition. Internationally, concepts such as the "planetary village"[6], "ecovillage"[7] and "sustainable village"[8] have all laid the foundation for the development of low-carbon villages. Compared with sustainable villages, low-carbon villages do not encompass every social, political and economic dimension, but instead approach the issue from a planning perspective, seeking solutions that can be implemented in space; compared with ecological villages, low-carbon villages do not focus solely on the restoration of ecological resources, but also consider improvements in the quality of the living environment and people's living standards.

In terms of theoretical research, carbon cycle theory provides a basic framework for the study of low-carbon villages[9]. Activities in villages such as agricultural production, energy consumption and transportation form carbon sources, while ecosystems such as forests, farmland and water bodies form carbon sinks. Meanwhile, urban metabolism theory emphasizes that there are exchanges of material and energy flows between rural and urban areas[10]; villages not only perform production functions but also undertake functions of resource recycling and ecological services.

Drawing on UN-Habitat's Net-Zero Carbon Village Planning Guidelines, this study defines a low-carbon village as a sustainable, low-carbon human settlement system achieved through a comprehensive approach that takes into account climatic conditions, land use, the local economy, renewable energy utilization, zero pollution, the water cycle, biodiversity and transportation. Low-carbon rural construction is not a results-oriented assessment of carbon emissions, but a long-term and dynamic process. This study focuses on which planning pathways should be used in this process, and divides the specific planning contents into six aspects: overall planning layout, industrial planning, transportation planning, energy and infrastructure planning, green space system planning, and architectural design guidelines.

2.2 Practical Cases of Low-Carbon Villages

International low-carbon village practices are mainly concentrated in Europe, the United States and the Asia-Pacific region, focusing on renewable energy utilization, smart agriculture and other localized strategies.

The Findhorn Ecovillage in Europe is a typical ecovillage case. It achieves partial energy self-sufficiency through wind, biomass and solar energy, adopts biotechnology for sewage treatment, and develops environmentally friendly and energy-efficient architectural design methods to reduce energy consumption and environmental impacts. The village has also developed organic agriculture, increasing the supply of local agricultural products, improving food quality, and reducing carbon emissions from transportation.

The Abundance EcoVillage in the United States emphasizes ecological circular agriculture and resource reuse. The village is powered by wind and solar energy, and has built a rainwater harvesting system and a constructed wetland sewage treatment system, while emphasizing ecological landscape construction and low-energy building design. Local houses significantly reduce energy consumption through shading and natural lighting, and average household energy consumption is far below the local average.

The Konohana Family in Japan embodies a development model that combines traditional agriculture with modern technology. The village practices organic and fertilizer-free farming, uses solar, geothermal and biomass energy for its energy supply, and treats waste through a composting system. In terms of architecture, it emphasizes the use of natural materials and the functional adaptation of traditional buildings, while introducing energy-saving equipment and energy management systems.

Overall, international low-carbon village cases show certain commonalities in terms of industry, energy and infrastructure, green space and architecture: first, explorations are centered on agriculture; second, local clean energy is actively utilized and developed, with some cases emphasizing the ecological treatment of sewage, garbage and waste; third, attention is paid to waterfront greening; and fourth, emphasis is placed on guiding the renovation and design of buildings. These planning experiences provide references for low-carbon rural construction in China.

2.3 Planning Cognition of Low-Carbon Villages

The goals of low-carbon rural construction include reducing carbon emissions and increasing carbon sinks. The former requires reducing the total demand for energy consumption and adjusting the structure of energy supply[11], while the latter requires enhancing the carbon absorption of natural resources. In terms of spatial levels, low-carbon rural construction can be divided into three levels: the village territory scale, the village settlement scale and the building scale. The village territory includes not only the settlement but also resources such as farmland, woodland and water sources, and involves the overall land use layout and transportation planning; the village settlement is both the aggregation of the low-carbon effects of architectural spaces and a reflection of the low-carbon nature of land use structure and layout at a larger scale[12], involving overall planning layout, industrial planning, transportation planning, energy and infrastructure planning, green space system planning and architectural design guidelines; the individual building is the basic unit of rural planning and the object of architectural design, as well as the spatial level most closely linked to residents' daily life, lifestyles and equipment use; it is directly related to architectural design guidelines and indirectly affects industrial planning as well as energy and infrastructure planning.

With reference to existing standard systems for low-carbon rural construction in China and abroad, the key planning points for low-carbon rural construction are listed. Explanations in existing planning studies of how each key point achieves carbon reduction goals demonstrate the rationality of the listed points, while descriptions in practical cases of how each key point works in specific villages demonstrate their feasibility. The key planning points are divided into six aspects—overall planning layout, industrial planning, transportation planning, energy and infrastructure planning, green space system planning, and architectural design guidelines—which cover the main components of low-carbon rural construction and can effectively achieve low-carbon goals.

First, overall planning layout. The overall planning layout of a low-carbon village takes a macro perspective, focusing on the coordination among land use layout, spatial texture and the external natural environment. Different land use types act as carbon sources and sinks of varying intensities, and the regulation of land use changes the intensity and combination of carbon sources and sinks[13]. Specifically, it can be shaped through two aspects: an energy-efficient spatial structure and a compact, mixed land use layout.

Second, industrial planning. The overall approach of industrial planning for low-carbon villages is to change the existing industrial structure or to achieve transformation and upgrading within a particular industry. Changing the industrial structure means choosing to promote, retain or phase out industries according to the specific conditions of the village, while transformation within an industry mainly targets traditional agriculture and manufacturing. Specifically, it can be divided into three aspects: increasing the share of tourism, energy conservation and emission reduction in manufacturing, and the smart and ecological transformation of agriculture.

Third, transportation planning. Transportation planning for low-carbon villages aims to reduce carbon emissions generated by the travel of villagers and tourists, mainly by reducing the use of private motor vehicles. Specifically, it can be shaped through three aspects: building a slow traffic system, improving the public transportation system, and introducing low-carbon vehicles.

Fourth, energy and infrastructure planning. The energy and infrastructure system of a low-carbon village mainly refers to the energy utilization system, the water supply and drainage system, and the garbage and excrement treatment system. The provision and management of various energy sources and resources constitute the core support for the daily production and life of rural residents, and are also key factors affecting carbon emissions. Therefore, clean, efficient and sustainable energy and infrastructure planning is one of the priorities of low-carbon rural construction.

Fifth, green space system planning. Compared with cities, villages are closer to nature and have inherent ecological advantages. The design of green spaces can not only provide visual enjoyment for villagers and tourists, but also play landscape and ecological roles, regulating the microclimate and absorbing carbon dioxide. By building a multi-level green space system, the specific goal of increasing natural carbon sinks can be achieved.

Sixth, architectural design guidelines. This study seeks to coordinate the relationship between buildings and their surroundings and to make full use of the material resources left over from history, and on this basis proposes architectural design guidelines. This subsection mainly considers carbon emissions in the production and operation stages of buildings; carbon emissions in the demolition stage are included in waste treatment under energy and infrastructure planning.

Based on the above, the planning contents and key points of low-carbon rural construction are listed in Table *1*.

Table 1 Planning Contents and Key Points of Low-Carbon Rural Construction

Planning Content		**Key Planning Point**	**Carbon Reduction Principle**	**Spatial Level**	**Reference Standard**
Overall planning layout		Energy-efficient spatial structure	Improving land use efficiency[14]	Village territory	Guidelines for the Construction of Low-Carbon Community Pilots
		Compact and mixed land use layout	Shortening travel distances[11] Concentrated and efficient land use[15]	Village settlement	Low-Carbon City Promotion Act
Sectoral planning	Industrial planning	Increasing the share of tourism	Reducing the environmental impact of economic development[16]	Village settlement	Blue Book on Low-Carbon Rural Construction in Zhejiang Province
		Energy conservation and emission reduction in manufacturing	Improving industrial energy efficiency[17]	Village settlement	
		Smart and ecological transformation of agriculture	Improving resource efficiency through efficient production[18] Promoting resource recycling and economic development[19]	Village territory	
	Transportation planning	Building a slow traffic system	Reducing dependence on motorized transport[16]	Village territory, village settlement	Low-Carbon City Promotion Act
		Encouraging public transportation	Reducing private vehicle use[20]	Village settlement	
		Introducing low-carbon vehicles	Adjusting the energy structure of vehicles[21]	Village territory	
	Energy and infrastructure planning	Renewable energy utilization	Reducing greenhouse gases generated by energy use[22]	Village territory, building	Low-Carbon City Promotion Act
		Resource recycling	Turning waste into resources[23]	Village settlement	European Green Deal
	Green space system planning	Improving public green space design	Enhancing plant carbon sinks and improving the microclimate[24]	Village settlement	Low-Carbon City Promotion Act
		Making full use of attached green space		Building	
Architectural design guidelines		Energy-saving retrofit of existing buildings	Reducing energy for the production, transport and operation of building materials[25]	Building	Low-Carbon City Promotion Act
		Energy-efficient design of new buildings	Improving thermal performance and reducing lighting	Building	Guidelines for the Construction of Low-Carbon

Planning Content	Key Planning Point	Carbon Reduction Principle	Spatial Level	Reference Standard
		demand[26]		Community Pilots

(Source: compiled by the author)

3 Empirical Study: Three Typical Villages in Zhejiang Province

3.1 Basic Information on the Case Villages

Zhejiang Province has long been at the forefront of rural construction in China in terms of both awareness and practice, and can therefore serve as a model for study. This study selects Xinjian Village in Ganlan Town, Dinghai District, Zhoushan City; the Lotus International Future Community in Lotus Town, Qujiang District, Quzhou City; and Shatan Village in Yutou Township, Huangyan District, Taizhou City as typical cases. All three villages began low-carbon rural construction relatively early, and have become low-carbon pilot sites designated in government documents, case sites in standards and guidelines, or award winners in international sustainability competitions. The locations of the three villages are shown in Figure 1.



Figure 1 Locations of the case villages in Zhejiang Province

(Source: drawn by the author)

The basic information on the three typical villages is as follows. Xinjian Village has successively been awarded titles such as “China’s Most Beautiful Village and Town”, “National Ecological Culture Village” and “National Beautiful and Livable Demonstration Village”, and was included in the Net-Zero Carbon Village Planning Guidelines: The Case of the Yangtze River

Delta Region of China; its achievements have been compiled into Rural Revitalization Practices in Dinghai District, Zhoushan under the Net-Zero Carbon Goal, giving it strong demonstrative value and significance for frontier exploration. The Lotus International Future Community is an important representative of Zhejiang Province's "Future Community" and "Rural Future Community" pilots; it was selected as a pilot unit of the UN Guidelines for Sustainable Cities and Communities and received a UN environmental sustainability award, reflecting a practical pathway for integrating low-carbon villages with future communities. Shatan Village takes the net-zero carbon goal as the direction of its ecological revitalization, and its old street block was chosen as the cover case of the Net-Zero Carbon Village Planning Guidelines: The Case of the Yangtze River Delta Region of China, giving it strong value for international dissemination and as a typical demonstration.

3.2 Analysis of Results

(1) Xinjian Village, Dinghai District, Zhoushan City

Xinjian Village is located in the southern part of Ganlan Town, Dinghai District, Zhoushan City. Nestled between mountains and the sea, the village has ecological resources such as forests, water bodies and farmland, and its overall ecological environment is good. Community income comes mainly from crop farming, homestays and other services, with the tertiary sector as the leading industry. In 2019, a unified carbon emission accounting was carried out for five villages in Dinghai District. The comparison shows that per capita carbon emissions in Xinjian Village were only 1.40 t CO2e per person, far below the average levels of Zhoushan City (8.63 t CO2e per person) and Zhejiang Province (8.89 t CO2e per person). The total carbon source emissions of Xinjian Village amounted to 1,688 t CO2e, total carbon sink absorption to −1,396 t CO2e, and total net emissions to 3,225 t CO2e. Residents' household carbon emissions mainly come from domestic electricity use and travel; agricultural carbon emissions come entirely from crop cultivation; and rural industrial carbon emissions mainly come from homestay operations.

In terms of overall planning layout, Xinjian Village has achieved an energy-efficient spatial structure and a compact, mixed land use layout. In terms of industrial planning, it has developed handicrafts and sightseeing tourism centered on marine culture. In terms of transportation planning, it has built the first national mountaineering and fitness trail on an island, reducing transport-related carbon emissions by creating a slow traffic system. In terms of energy and infrastructure

planning, as a UN net-zero carbon village demonstration base, Xinjian Village has experimented with a variety of clean energy equipment, including solar, wind and biomass energy. However, during formal implementation it was found that villagers' acceptance of clean energy was low owing to concerns such as damage to the village's appearance and noise pollution, so such equipment has not yet been widely promoted in scale or scope, which shows the importance of raising ecological awareness. The village's planning team has responded promptly by using new-type tiles to maintain visual consistency; whether this can dispel villagers' concerns, as well as its cost-effectiveness, remains to be tested in the subsequent rollout. In addition, Xinjian Village has made advanced attempts in infrastructure, introducing cutting-edge technologies such as smart infrastructure and ecological sewage treatment ponds. In terms of green space system planning, Xinjian Village has well preserved its existing ecological resources and has renovated, restored and designed its green spaces and water bodies, which helps improve the rural landscape and supports the development of sightseeing tourism through marine culture. In terms of architectural design guidelines, Xinjian Village is the only one of the three villages that still has large-scale construction projects under way; the formulation, implementation and supervision of architectural design guidelines need to be gradually improved in future construction.

Overall, low-carbon rural construction in Xinjian Village emphasizes the precise calculation of carbon emissions under the dual carbon goals of the new era and the exploration of many advanced technical means. It has strong leading and reference value and has provided inspiration for low-carbon rural construction in other parts of Zhejiang Province.

(2) Lotus International Future Community, Qujiang District, Quzhou City

The Lotus International Future Community is located on the plain in the central part of Qujiang District, adjacent to airport logistics facilities and with convenient transportation. Agriculture and tourism have developed through the construction of the Lotus Modern Agricultural Park.

In terms of overall planning layout, the Lotus Future Community currently suffers from relatively fragmented land use, which is not sufficiently intensive or efficient. In terms of industrial planning, the Lotus Future Community leads the three cases in smart and industrialized development; it has introduced real-time monitoring and result-visualization equipment in infrastructure, industry and other fields, and has actively promoted integrated development of the

primary, secondary and tertiary sectors to foster a circular economy, thereby improving service efficiency and resource efficiency and reducing carbon emissions in a smart and circular way. In terms of energy and infrastructure planning, the Lotus Future Community was the first of the three cases to install photovoltaic panels on villagers' houses, illustrating the importance of a clear profit scheme and a convenient installation procedure; meanwhile, the failure to effectively implement agrivoltaics shows that the relationship between crop yields and solar power generation must be considered, and its implementation experience is instructive. In terms of green space system planning, the Lotus Future Community emphasizes combining agriculture with sightseeing and combining waterscape creation with sponge system construction, improving ecosystem performance and carbon sink capacity through resource recycling and industrial chain integration. In terms of architectural design guidelines, current architectural design still focuses mainly on appearance and lacks systematic guidelines, which require continuous improvement in subsequent construction.

Overall, as a pilot of Zhejiang's future communities in rural areas, the Lotus Future Community has integrated the development of the Zhixi Stream and the Shengshi Lotus Agricultural Sightseeing Park through smart technology and industrial linkage, and has developed relatively mature operating procedures and promotion mechanisms. It offers strong lessons in planning implementation, circular economy and green space system planning.

(3) Shatan Village, Huangyan District, Taizhou City

Shatan Village is located in the southeastern part of Yutou Township, Huangyan District, Taizhou City, Zhejiang Province. The village territory is mountainous but has good transportation conditions, and it retains a relatively intact traditional street pattern and historic buildings. The village is based on agriculture and animal husbandry, and develops rural tourism by combining natural landscapes with traditional streets.

In terms of overall planning layout, the planning and design of Shatan Village made maximum use of natural resources and environmental advantages, achieving efficient and rational land use. In terms of industrial planning, Shatan Village has combined traditional cultural resources with linkages to scenic spots in neighboring villages and towns, steering its industrial structure toward low environmental impact and high economic returns; the development of tourism has also created jobs and raised the incomes of villagers and the village collective, making the entire construction

process more sustainable. In terms of transportation planning, Shatan Village has reduced transport-related carbon emissions by building a slow traffic system and encouraging public transportation. In terms of energy and infrastructure, Shatan Village has made full use of traditional biomass energy, explored the possibility of biogas power generation, and launched pilot projects for new energy sources such as solar power, providing diversified energy sources. Although the projects are small in scale, their siting was carefully conceived and their monitoring data are detailed and real-time; however, given that Shatan Village is a traditional village, photovoltaic equipment is not suitable for large-scale promotion. Shatan Village has also achieved the recycling of water and solid waste resources, indirectly reducing resource demand as well as carbon emissions from production and transportation. In terms of green space system planning, it has regulated the wind and thermal environment of the whole village, enhanced living comfort, created a pleasant living experience, and helped strengthen the resilience of the rural ecosystem and give full play to its carbon sink capacity. In terms of architectural design guidelines, Shatan Village has reused idle buildings, improved resource efficiency, and introduced appropriate technologies, reducing active energy use.

Overall, the low-carbon rural planning approach of Shatan Village mainly reflects the full use of existing resources, the integrated introduction of appropriate technologies and the pleasant design of the street environment. It is a distillation of the natural wisdom of rural construction in China; its economic and environmental costs are relatively low, and it is highly operable and instructive.

3.3 Comparison and Results

The three villages differ in economic development, population size and land area (see Table 2). Xinjian Village has a moderate land area and population and the highest per capita income; its village collective has a considerable economic foundation and is in a position to explore advanced technologies. The Lotus Future Community has the largest land area, the largest population and a middle level of per capita income; in addition to the administrative village, the Lotus Modern Agricultural Sightseeing Park and the Zhixi Stream are included in the core pilot construction area, so it is suited to exploring a low-carbon rural construction pathway centered on industry and resource recycling. Shatan Village has the smallest land area, the smallest population and the lowest per capita income; its advantage lies in its profound historical and cultural heritage, so rather than introducing too many high technologies that do not match its current economic

development, it should make full use of its existing strengths.

Table 2 Basic Information on the Case Villages

	Xinjian Village, Dinghai, Zhoushan	**Lotus Future Community, Qujiang, Quzhou**	**Shatan Village, Huangyan, Taizhou**
Area	450ha	800ha	200ha
Population	1,207	5,874	1,112
Per capita income	CNY 45,100/year	CNY 30,000/year	CNY 20,000/year
History and culture	Historical verification shows that Nandong'ao was recorded as early as the Ganlan Village map of the Guangxu reign of the Qing Dynasty; the village features natural ecological charm, rural farming life and folk customs unique to the island.	With a history of over a hundred years, it has five types of distinctive culture: ancient architecture and intangible cultural heritage, farming culture, lotus culture, local-worthy and neighborhood culture, and integrity culture.	According to existing historical records, Shatan Village has a history of nearly 900 years and has nurtured Taoist culture, Confucian culture, farming culture and architectural culture.
Topography	The village lies in a shallow valley in a mountainous area. Located in the subtropics on the Pacific coast and adjoining the Eurasian continent, it has cold winters and hot summers, abundant rainfall throughout the year, strong winds and ample sunshine.	The terrain is relatively flat, and overall it belongs to hilly and mountainous terrain. Sunlight is good and abundant; the climate is subtropical monsoon, with hot and humid summers, cold and humid winters, and mild springs and autumns. Precipitation is abundant.	Hilly and mountainous landforms with a subtropical monsoon climate and favorable climatic conditions. The prevailing winds are southeasterly and northeasterly; typhoons are frequent at the turn of summer and autumn, accompanied by rainstorms that often cause floods.
Socio-economy	Community income comes mainly from crop farming (beans, fruit, vegetables and oil crops), homestays and other services; the village is transitioning from the primary to the tertiary sector.	The main pillar industries are grain production and animal husbandry.	The leading industries of Shatan Village are rice and fruit cultivation and animal husbandry. Tourism has flourished in recent years.

The planning pathways of the three villages are then compared (see Table 3).

Table 3 Comparison of Planning Pathways for Low-Carbon Rural Construction in the Three Typical Case Villages

Planning Content	**Key Planning Point**	**Planning Pathway**	**Xinjian Village, Dinghai, Zhoushan**	**Lotus Future Community, Qujiang, Quzhou**	**Shatan Village, Huangyan, Taizhou**
Overall planning layout	Energy-efficient spatial structure	Suitability-based territorial spatial development	√	--	√
		Spatial texture conforming to	--	--	√

Planning Content		Key Planning Point	Planning Pathway	Xinjian Village, Dinghai, Zhoushan	Lotus Future Community, Qujiang, Quzhou	Shatan Village, Huangyan, Taizhou
			landscape and terrain			
		Compact and mixed land use layout	Creating a 15-minute rural community life circle	√	√	√
			Compact and intensive land use	√	--	--
Sectoral planning	Industrial planning	Increasing the share of tourism	Exploring distinctive tourism resources	√	--	√
		Energy conservation and emission reduction in manufacturing	Adjusting the structure of manufacturing	--	--	√
		Smart and ecological transformation of agriculture	Strengthening technical support for smart agriculture	--	√	--
			Developing ecological circular agriculture	--	√	--
	Transportation planning	Building a slow traffic system	Providing comfortable cycling and walking routes	√	√	√
			Creating a pleasant street environment	--	--	√
		Encouraging public transportation	Deploying shared bicycles	--	√	√
			Deploying sightseeing vehicles	√	--	--
		Introducing low-carbon vehicles	Siting EV charging stations	√	√	√
	Energy and infrastructure planning	Renewable energy utilization	Layout of photovoltaic equipment	√	√	√
			Layout of other clean energy generation equipment	√	--	√
		Resource recycling	Rainwater and reclaimed water use	√	√	√
			Sorted collection and treatment of garbage	√	√	√

Planning Content		Key Planning Point	Planning Pathway	Xinjian Village, Dinghai, Zhoushan	Lotus Future Community, Qujiang, Quzhou	Shatan Village, Huangyan, Taizhou
			Resource utilization of excrement	--	√	√
	Green space system planning	Improving public green space design	Optimizing park and green space design	√	√	--
			Creating waterfront ecological greening	√	√	√
			Promoting agricultural sightseeing greening	√	√	√
		Making full use of attached green space	Adding courtyard and street-corner greening	--	√	--
Architectural design guidelines		Energy-saving retrofit of existing buildings	Reuse of idle buildings	√	√	√
			Introduction of appropriate technologies and energy-saving equipment	√	--	√
		Energy-efficient design of new buildings	Optimizing building form and orientation	--	--	√
			Using local materials	--	√	√

Note: √ indicates measures observed as implemented during the field survey; -- indicates measures not yet completed.

(Source: compiled by the author)

The demonstrative value of Xinjian Village lies mainly in energy and infrastructure construction. Making full use of the abundant solar and wind resources of Dinghai District, Zhoushan, the village took the lead in applying advanced technologies such as integrated wind–solar–storage equipment and power generation monitoring and visualization to low-carbon rural construction, and has upgraded its photovoltaic tiles to a new version that does not affect the village's appearance. Meanwhile, Xinjian Village converted a former swimming pool and used a wetland purification method to achieve the ecological treatment and recycling of sewage. It has adjusted its energy structure through clean energy generation and introduced ecological infrastructure to beautify the landscape and reduce negative environmental impacts. In addition,

Xinjian Village was the first to achieve precise carbon emission accounting, providing quantitative support for the effectiveness of the planning pathways.

The demonstrative value of the Lotus Future Community lies in green space and industry. The community has made full use of existing wetland resources to create a multi-level green space system, including an ecological axis along the Zhixi Stream, agricultural park sightseeing, a community wetland park and "one-meter vegetable garden" courtyard greening, and has applied sponge city technologies to increase permeability and promote the water cycle. In terms of industry, the community has actively developed a modern agricultural park, optimizing crop growing conditions with smart agriculture technologies, and has vigorously promoted the linked development of agriculture and other industries, extending the industrial chain, adding value at each link and promoting the circulation of resources and production factors, thereby achieving synergistic ecological and economic benefits.

The demonstrative value of Shatan Village lies mainly in architecture and transportation. In terms of architecture, it has effectively reused formerly abandoned buildings, "turning waste into treasure". These buildings are structurally well preserved and located in the village center, which facilitates the creation of a 15-minute life circle, and their property rights belong to the government, making implementation highly feasible. The renovation is not limited to improving appearance and repairing façades, but also emphasizes modern means such as energy-saving technologies and high-efficiency equipment. In terms of transportation, Shatan Village has carefully designed the pedestrian environment of Rouji Street and developed hiking and cycling routes in conjunction with surrounding villages, creating a comfortable slow traffic system that both showcases the rural landscape and effectively reduces carbon emissions from motor vehicles.

Despite their different development paths, the three villages share commonalities in low-carbon rural construction. First, all three villages attach importance to the low-carbon construction of rural space, implementing the requirements of ecological and environmental carrying capacity and territorial development suitability under territorial spatial planning, achieving 15-minute accessibility of public service facilities and mixed land use, thereby using land efficiently, avoiding waste of resources and giving full play to the functions of ecological and agricultural space. Second, all three villages need to develop sustainable tourism. The Rouchuan Scenic Area in Shatan Village and the Nandong Art Valley in Xinjian Village are both national 4A-level scenic areas, and the Lotus Future Community also intends to develop tourism in conjunction with agriculture; all three

villages face the common challenge of reducing carbon emissions from tourists and the tourism industry.

4 Recommendations on Planning Pathways for Low-Carbon Rural Construction

4.1 Overall Planning Layout

(1) Energy-efficient spatial structure

Suitability-based territorial spatial development requires a comprehensive assessment of environmental and ecological conditions to determine the most suitable land use types and boundaries, delineate binding indicators, and, where conditions permit, calculate the carbon emissions of each land use type. A spatial texture that conforms to the landscape and terrain requires that village siting and layout fully consider natural resources such as geology and landforms, hydrology and vegetation, sunlight and ventilation corridors, drawing on the ancestral siting principle of backing onto mountains and facing water, so as to achieve harmonious coexistence between humans and nature.

(2) Compact and mixed land use layout

Create a 15-minute rural life circle and promote mixed-use development to ensure the coexistence of residential, commercial, educational, recreational and medical facilities, and encourage one-stop public service stations. Intensive and compact land use requires consolidating fragmented land, coordinating rural land transfer and the reclamation of abandoned industrial and mining land, and rationally planning residential, industrial and agricultural areas to avoid disorderly expansion and waste of land resources.

4.2 Industrial Planning

(1) Increasing the share of tourism

Actively develop rural services with low environmental impact, explore and protect local resources such as traditional handicrafts, folk activities and natural scenery, build sound infrastructure, strengthen community participation and training, and carry out effective marketing. This approach is suitable for villages rich in historical and cultural or natural landscape resources.

(2) Energy conservation and emission reduction in manufacturing

Promote energy conservation and emission reduction in rural manufacturing by adjusting the

industrial structure and optimizing production modes. Minimize energy-intensive and highly polluting industries, optimize enterprises that must be retained, build industrial parks to reduce transport energy consumption through energy sharing, or improve production processes to promote clean, electrified and intelligent upgrading.

(3) Smart and ecological transformation of agriculture

Monitor and control the entire agricultural production process in a refined manner, and reduce excessive fertilization and irrigation through automatic irrigation systems, drone spraying and smart greenhouses. Develop ecological circular agriculture by using livestock manure as fertilizer for crops and crop straw as feed for livestock. Promote agritourism such as fruit picking and farm sightseeing.

4.3 Transportation Planning

(1) Building a slow traffic system

Provide cycling lanes and walking routes integrated with tourism projects in the village territory, using paving materials that balance safety and comfort. Create a pleasant street environment: when planning and designing well-preserved old streets, adopt local materials and architectural styles, and, together with control measures that restrict private cars from entering core scenic areas, encourage tourists to use slow modes of travel.

(2) Encouraging public transportation

Deploy shared bicycles, placing rental points in line with the layout of tourist attractions and rationally planning their locations. Deploy sightseeing vehicles, planning routes and stops in line with tourist routes, and regularly evaluate route efficiency and visitor satisfaction.

(3) Introducing low-carbon vehicles

Install EV charging facilities, taking into account the convenience and accessibility of locations, covering major traffic routes and public places, ensuring adequate supporting power facilities, and providing easy-to-use payment and service systems.

4.4 Energy and Infrastructure Planning

(1) Renewable energy utilization

Photovoltaic equipment can be installed on roofs, carports and street lights, taking into account sunlight conditions and the wishes of the parties concerned, but it should be promoted with caution in traditional villages with landscape conservation requirements. Other clean energy generation equipment includes small wind turbines in areas with higher wind speeds, small dams or hydro turbines in areas with abundant water resources, and anaerobic digestion of straw and animal manure to generate energy.

(2) Resource recycling

Rainwater and reclaimed water use requires coordinating rainwater harvesting with sewage treatment: rainwater is treated through filters, sedimentation tanks and disinfection facilities, and sewage treatment shifts toward ecological treatment processes. Sorted collection and treatment of garbage requires hazardous waste to be sealed and stored centrally, recyclables to be recovered, leftover food and straw to be used as livestock feed, and construction waste to be sorted and crushed. Resource utilization of excrement requires building or renovating ecological toilets and smart toilets, and using excrement for power generation and soil fertility improvement through urine diversion, biogas fermentation and other means.

4.5 Green Space System Planning

(1) Improving public green space design

Optimize the design of parks and green spaces, improve planting configurations, and choose path materials such as gravel and wooden boards. Promote agricultural sightseeing greening, preserving the distinctive regional pastoral landscape and creating spatial layouts that serve both agricultural production and tourism and leisure. Create waterfront ecological greening, using rural water bodies such as ponds, streams and canals to regulate the microclimate, rebuilding ecosystems around natural water bodies, and stabilizing banks with natural materials and vegetation belts.

(2) Making full use of attached green space

Plant greenery at wall corners and in vacant spaces beside houses, filling every available gap, and encourage villagers to grow ornamental plants or seasonal fruit and vegetables in their own courtyards; where conditions permit, provide green roofs on individual buildings or vertical greening with climbing plants.

4.6 Architectural Design Guidelines

(1) Energy-saving retrofit of existing buildings

Abandoned buildings that are structurally sound and have regional characteristics can be given modern functions; implementation requires an assessment of building quality, location and property rights. The introduction of appropriate technologies and energy-saving equipment requires considering different materials for roofs, walls, doors and windows, and floors, using energy-saving equipment such as heat recovery ventilators and air-source heat pump water heaters, and introducing technologies such as smart lighting and smart paper dispensers in public buildings and spaces.

(2) Energy-efficient design of new buildings

Optimize building form and orientation to facilitate natural ventilation and daylighting, and choose row-house and stacked building configurations. Use local materials such as stone, timber, bamboo and straw for exterior walls and interior partitions to maintain the landscape character and rural atmosphere.

5 Conclusion

Against the background of the national dual carbon goals and the rural revitalization strategy, this study explores locally adapted and operable planning pathways for low-carbon rural construction. It constructs a planning cognition framework aimed at achieving low-carbon rural construction, distills planning and design methods and implementation pathways in six aspects, and conducts field research on Xinjian Village in Ganlan Town, Dinghai District, Zhoushan City; the Lotus International Future Community in Lotus Town, Qujiang District, Quzhou City; and Shatan Village in Yutou Township, Huangyan District, Taizhou City. The outcomes of existing low-carbon rural construction practices show that low-carbon rural construction is incremental: it requires dynamic adjustment based on positive and negative feedback during implementation, and should be advanced step by step through raising villagers' awareness, investing funds and policy support, and improving construction and operation models.

Low-carbon villages embody the new-era philosophy of high-quality development, high-quality living and high-efficiency governance, and reflect both thinking on climate issues and a response to the Sustainable Development Goals. By distilling recommendations on planning pathways for low-carbon rural construction from real cases, this study provides theoretical

reference and practical guidance for the vast number of rural construction practices across China.

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